Akoizo Original Practice - Form 5

Division B | Experimental Design | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Which is quantitative evidence?

Unit: Claims, evidence, and reasoning for trends and outliers.
Worked-case data: At levels 1,2,3, means are 12,16,20 units, with SD 2 at each level. Write a bounded trend CER.

A. Means increased from 12 to 20 units

B. It looks nice

C. The hypothesis is obviously true

D. The team worked hard

Answer: _____

2. (2 points) Randomizing order mainly helps address…

Unit: Problem, hypothesis, variables, and experimentation.
Worked-case data: All low-level trials occur early and all high-level trials occur late while the instrument drifts upward.

A. Every possible confounder automatically

B. Time-linked systematic differences

C. All measurement uncertainty

D. The need for units

Answer: _____

3. (4 points) Explain the evidence supporting your answer, including the calculation or comparison.

Unit: A clear Division B investigation report.
Worked-case data: 25 cm^2 towel: Absorbed masses: 2, 3, 4 g. 50 cm^2 towel: Absorbed masses: 5, 6, 7 g. Controls: Same towel brand, soak time, and draining method.
Separate investigation observations:
25 cm^2 towel: Absorbed masses: 2, 3, 4 g.
50 cm^2 towel: Absorbed masses: 5, 6, 7 g.
Controls: Same towel brand, soak time, and draining method.

Answer: _____
_____
_____

4. (2 points) High R^2 proves causation?

Unit: Quantitative data, graphs, and statistics.
Worked-case data: For response values 8, 10, 12, calculate mean, range, and sample SD.

A. No

B. Yes

C. Only with a colored line

D. Only when n = 3

Answer: _____

5. (4 points) Name the example independent and dependent variables.

Unit: The experiment and the prompt packet.
Worked-case data: A prompt supplies a ramp, object, ruler, and timer. Turn “study motion” into a testable question.

Answer: _____
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_____

6. (4 points) Why do more repeats not remove a fixed offset?

Unit: Errors, conclusions, and recommendations.
Worked-case data: Every timer reading is 0.5 s too high. Would ten more repeats eliminate the bias?

Answer: _____
_____
_____

7. (4 points) Why separate raw and summary data?

Unit: Materials, reproducible procedures, and qualitative data.
Worked-case data: A trial has no timer reading because the observer missed the endpoint. What should the table and procedure record?

Answer: _____
_____
_____

8. (4 points) Why keep raw data?

Unit: The experiment and the prompt packet.
Worked-case data: A prompt supplies a ramp, object, ruler, and timer. Turn “study motion” into a testable question.

Answer: _____
_____
_____

9. (4 points) Recommend a targeted improvement for reaction-time variation.

Unit: Errors, conclusions, and recommendations.
Worked-case data: Every timer reading is 0.5 s too high. Would ten more repeats eliminate the bias?

Answer: _____
_____
_____

10. (6 points) Solve this case and explain your reasoning: A trial has no timer reading because the observer missed the endpoint. What should the table and procedure record?

Answer: _____
_____
_____

11. (4 points) Give a targeted follow-up for excessive setup time.

Unit: Practice, timing, teamwork, and final review.
Worked-case data: A team finishes data collection with little time left and no graph or analysis. Suggest a targeted practice change.

Answer: _____
_____
_____

12. (6 points) Solve this case and explain your reasoning: At levels 1,2,3, means are 12,16,20 units, with SD 2 at each level. Write a bounded trend CER.

Answer: _____
_____
_____

13. (2 points) Define “Control condition” in the context of Experimental Design.

Answer: _____
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_____

14. (2 points) Define “Bottleneck” in the context of Experimental Design.

Answer: _____
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_____

15. (2 points) Define “Evidence” in the context of Experimental Design.

Answer: _____
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16. (2 points) Define “Protocol deviation” in the context of Experimental Design.

Answer: _____
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17. (2 points) Define “Pilot” in the context of Experimental Design.

Answer: _____
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18. (2 points) Define “Audit trail” in the context of Experimental Design.

Answer: _____
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